MAS-GPT: TRAINING LLMS TO BUILD LLM-BASED MULTI-AGENT SYSTEMS

AUTHORS: RUI YE, SHUO TANG, RUI GE, YAXIN DU, ZHENFEI YIN, SIHENG CHEN, JING SHAO

PRESENTED BY: MAHA ZAINAB

Agenda

Motivation

Research Contribution

Challenges

Methodology

Framework

Results

Strengths and Weaknesses

Extension

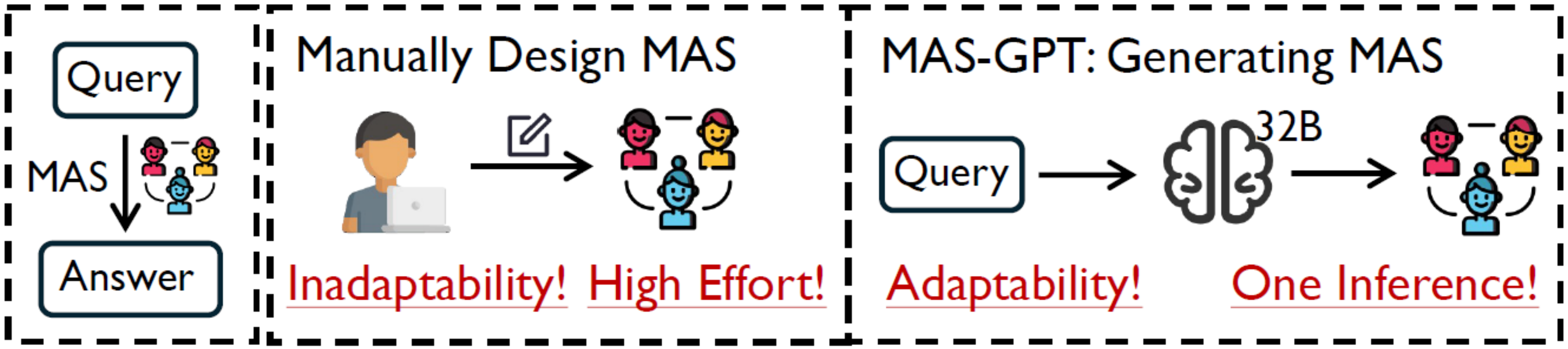


Figure 1: Introduction of our proposed new paradigm for building MAS. During inference, MAS-GPT adaptively generates a query-specific MAS with one LLM inference.

Motivation

- LLMs: Struggle with complex, multi-step, or domain-diverse tasks
- Multi-Agent Systems: Multiple LLMs to collaborate, improving reasoning and task decomposition.
- Manually Designed, Costly, and Static
- To automate MAS construction by enabling an LLM to generate executable multi-agent systems dynamically

Research Contributions

1. We reframe building MAS for each query as a generative language task. We unify the representation of MAS as executable code and propose a consistency-oriented query-MAS data construction pipeline for LLM training.

2. We introduce MAS-GPT, an LLM specifically trained to generate query-specific executable MAS. All code, data and models will be open-sourced.

3. Experiments on 9 benchmarks and 5 LLMs show that MAS-GPT consistently outperforms 10+ baselines, indicating its effectiveness, efficiency, and generalization ability.

Challenges

The limited knowledge of LLMs on the task of MAS generation and the lack of corresponding training data.

1. **How to represent the MAS**
2. **How to construct the dataset**

Methodology

User query: a multiagent system (MAS) is constructed

Multiple agents working collaboratively to generate the final answer.

```
def forward(query):
    math_agent = f'You are a math expert. Solve this\
        question: {query}'
    math_output = call_llm(math_agent)

    feedback_agent = f'Given {query} and {math_output},\
        provide feedback'
    feedback_output = call_llm(feedback_agent)

    refine_agent = f'Given {query}, {math_output} and \
        {feedback_output}, provide the final answer'

    return call_llm(refine_agent)
```

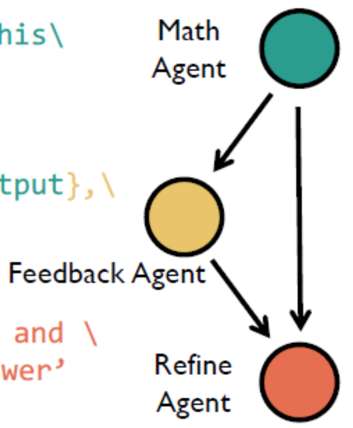


Figure 2: Our unified code representation of an executable MAS (i.e., a forward function). Each color denotes an agent. Agents defined by variables, LLM calls denoted by function calls, and interactions represented by string concatenations.

Framework

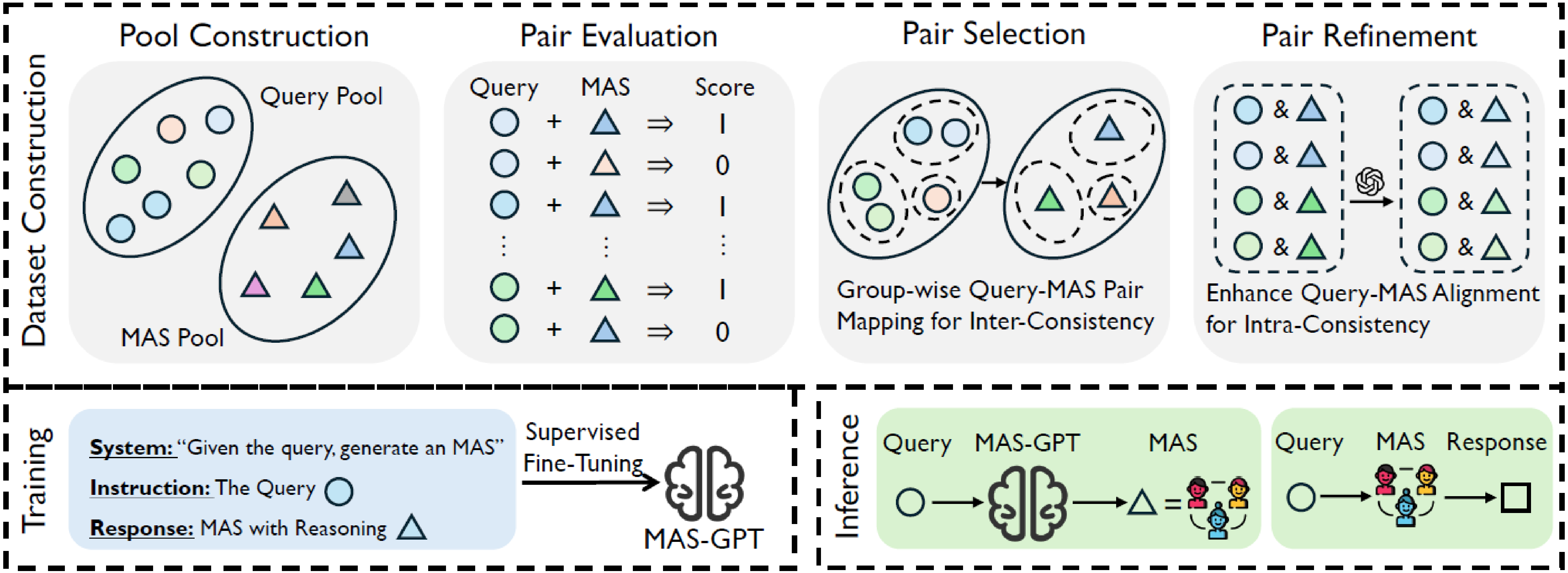
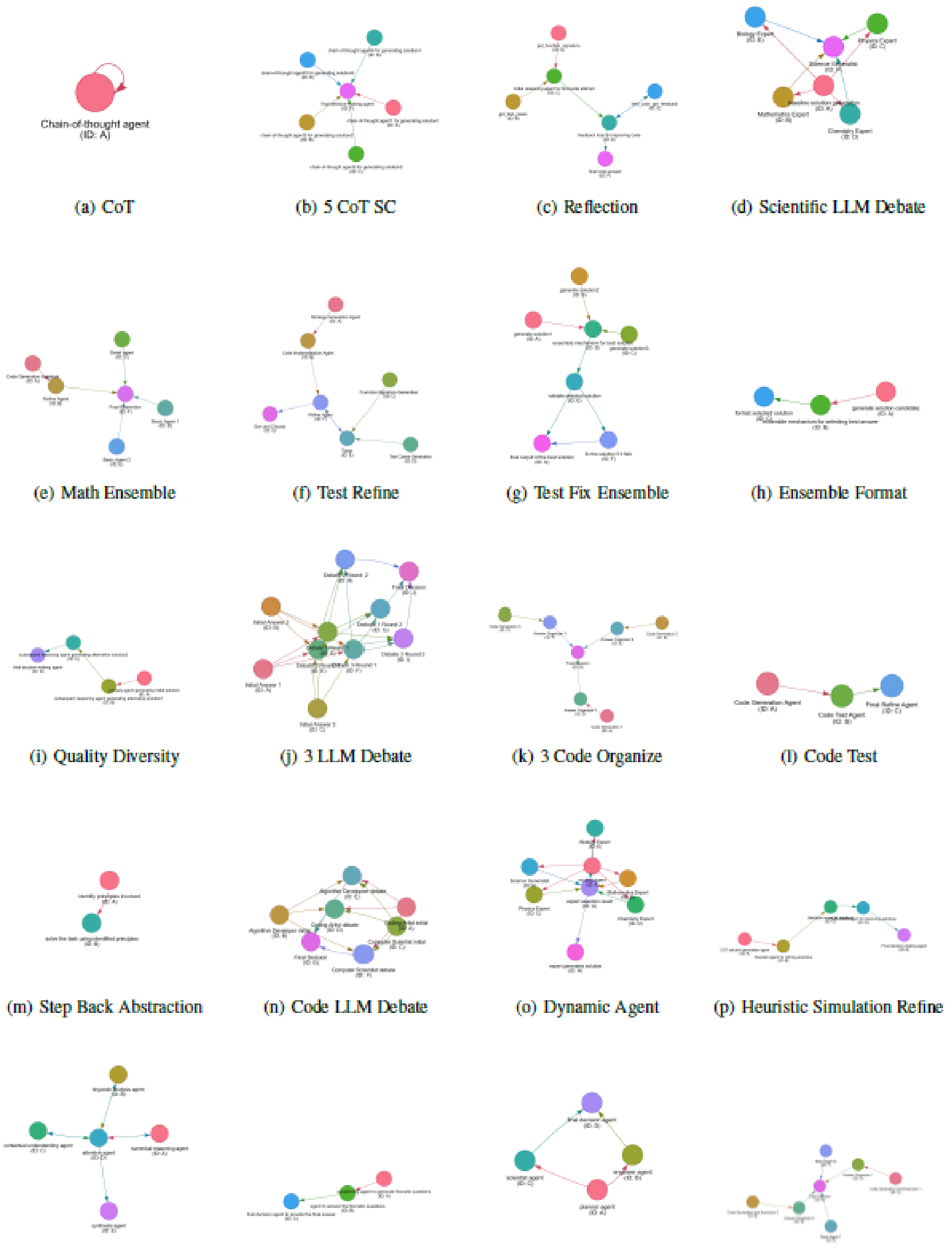


Figure 3: Illustrations of dataset construction, training, and inference of our proposed MAS-GPT.



(a) CoT (b) 5 CoT SC (c) Reflection (d) Scientific LLM Debate

(e) Math Ensemble (f) Test Refine (g) Test Fix Ensemble (h) Ensemble Format

(i) Quality Diversity (j) 3 LLM Debate (k) 3 Code Organize (l) Code Test

(m) Step Back Abstraction (n) Code LLM Debate (o) Dynamic Agent (p) Heuristic Simulation Refine

(q) Priority Refine (r) Socratic Questioning (s) Strategy Engineer—Scientist (t) 2 Code 2 Basic Ensemble

Figure 7: Visualization of our MAS pool

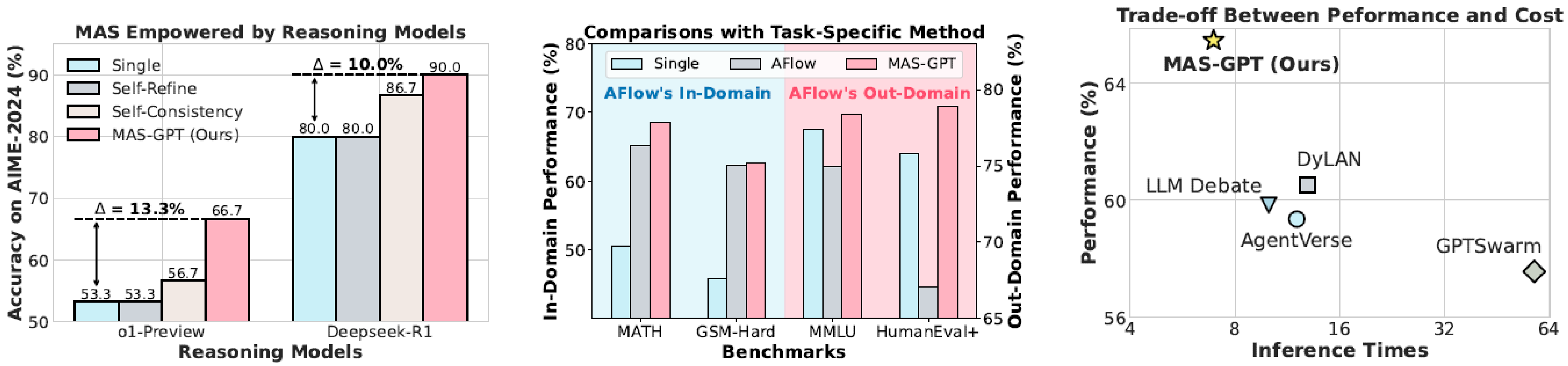
Results

MAS-GPT: Training LLMs to Build LLM-based Multi-Agent Systems

Table 2: Comparing MAS-GPT with 10 baselines across 8 benchmarks using Llama-3-70B-Instruct, MAS-GPT performs the best on average, verifying its generality in handling diverse queries. Benchmarks with * are out-of-domain for MAS-GPT.

Method	MATH	GSM8K	GSM-H	H-Eval*	H-Eval+*	MMLU	GPQA*	SciBench*	Avg.
Single (Dubey et al., 2024)	50.55	92.38	45.80	79.01	75.78	77.37	36.68	21.05	59.83
Chain-of-Thought (Wei et al., 2022)	53.20	92.79	46.20	77.16	77.02	75.56	35.28	17.68	59.36
Self-Consistency (Wang et al., 2024b)	61.59	94.99	47.20	77.78	75.78	78.18	37.15	20.00	61.58
LLM-Debate (Du et al., 2024)	61.37	91.58	44.60	74.69	74.53	77.78	34.35	19.79	59.84
Self-Refine (Madaan et al., 2024)	58.50	90.78	37.80	67.90	62.73	74.75	38.32	20.00	56.35
Quality-Diversity (Lu et al., 2024)	60.49	92.99	45.60	70.99	70.19	75.76	33.64	20.63	58.79
SPP (Wang et al., 2024c)	51.66	92.79	44.80	76.54	73.29	77.37	35.05	20.84	59.04
AgentVerse (Chen et al., 2024)	55.63	93.39	41.40	77.78	73.91	76.57	40.19	16.00	59.36
GPTSwarm (Zhuge et al., 2024)	55.41	93.19	43.20	69.14	73.91	75.15	36.45	14.11	57.57
DyLAN (Liu et al., 2024)	59.60	91.18	44.80	79.01	75.78	78.18	35.98	19.79	60.54
MAS-GPT (Ours)	68.65	93.39	62.40	80.25	78.88	78.38	37.62	24.21	**65.47**

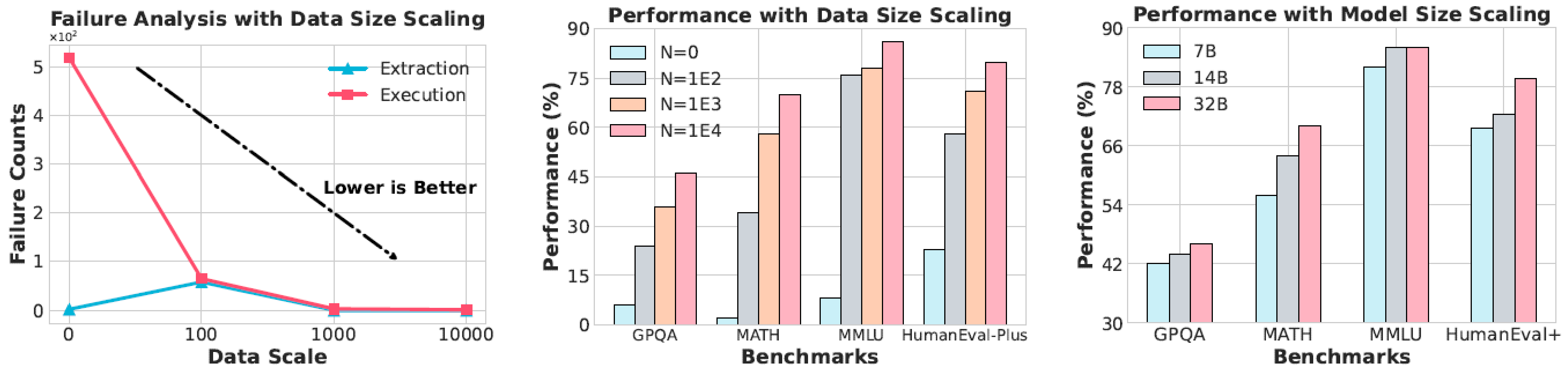
Results



(a) MAS-GPT-Assisted Reasoning (b) Comparison with Task-Specific Method (c) Performance v.s. Inference Times

Figure 4: (a) Different methods empowered with strong reasoning LLM: o1-preview. We see that our MAS-GPT significantly enhance the reasoning performance over single LLM, indicating its potential in further augmenting LLM reasoning. (b) Comparisons with AFlow (optimized on MATH). MAS-GPT even outperforms AFlow on its in-domain benchmarks; while AFlow fails on out-of-domain benchmarks. (c) MAS-GPT achieves the best performance with low inference cost.

Results



(a) Data Scale v.s. Failure (b) Data Scale v.s. Performance (c) Model Scale v.s. Performance

Figure 5: Explorations of scaling in training MAS-GPT. (a) More data leads to fewer execution failures. (b) More data contributes to better performance of MAS-GPT in facilitating MAS application. Without training (N=0), the model fails, highlighting that MAS generation is a non-trivial task requiring specific training. (c) Larger model generally contributes to better performance. These findings demonstrate the promising potential of MAS-GPT, suggesting that it can be further improved with more diverse, high-quality data and stronger models as the community continues to advance.

Generality

Table 3: MAS-GPT consistently performs the best across MAS-driving LLMs, indicating its strong compatibility.

METHOD	MATH	GSM-H	H-EVAL+	MMLU	GPQA	AVG.
QWEN2.5-72B-INSTRUCT						
SINGLE	85.86	64.91	85.37	82.60	44.39	72.63
COT	86.90	62.27	84.15	83.20	47.86	72.88
SELF-CON.	87.32	61.46	87.20	83.40	50.00	73.88
LLM-DEBATE	85.24	63.49	68.90	86.20	47.86	70.34
SELF-REFINE	83.58	59.03	78.66	85.40	43.32	70.00
Q-D	85.65	63.08	76.83	82.80	48.66	71.40
SPP	85.65	62.88	82.32	83.40	48.40	72.53
AGENTVERSE	84.82	59.43	81.10	83.20	44.65	70.64
GPTSWARM	83.16	63.89	83.54	84.60	44.92	72.02
DYLAN	87.73	63.08	85.37	84.40	51.07	74.33
MAS-GPT	87.53	66.33	85.98	83.80	48.66	**74.46**
GPT-4O-MINI-2024-07-18						
SINGLE	78.18	58.03	86.25	78.56	38.03	67.81
COT	78.79	60.84	85.62	79.16	39.60	68.80
SELF-CON.	81.62	59.04	85.00	80.96	39.82	69.29
LLM-DEBATE	79.60	60.84	86.25	80.76	37.81	69.05
SELF-REFINE	74.55	54.62	76.88	79.16	33.33	63.71
Q-D	79.80	59.64	84.38	79.76	37.58	68.23
SPP	77.58	57.63	86.25	77.96	37.58	67.40
AGENTVERSE	75.15	55.62	79.38	78.36	36.24	64.95
GPTSWARM	75.15	55.62	79.38	78.36	36.32	64.97
DYLAN	81.21	59.24	80.62	79.96	40.94	68.39
MAS-GPT	81.21	61.45	86.88	80.36	42.60	**70.50**

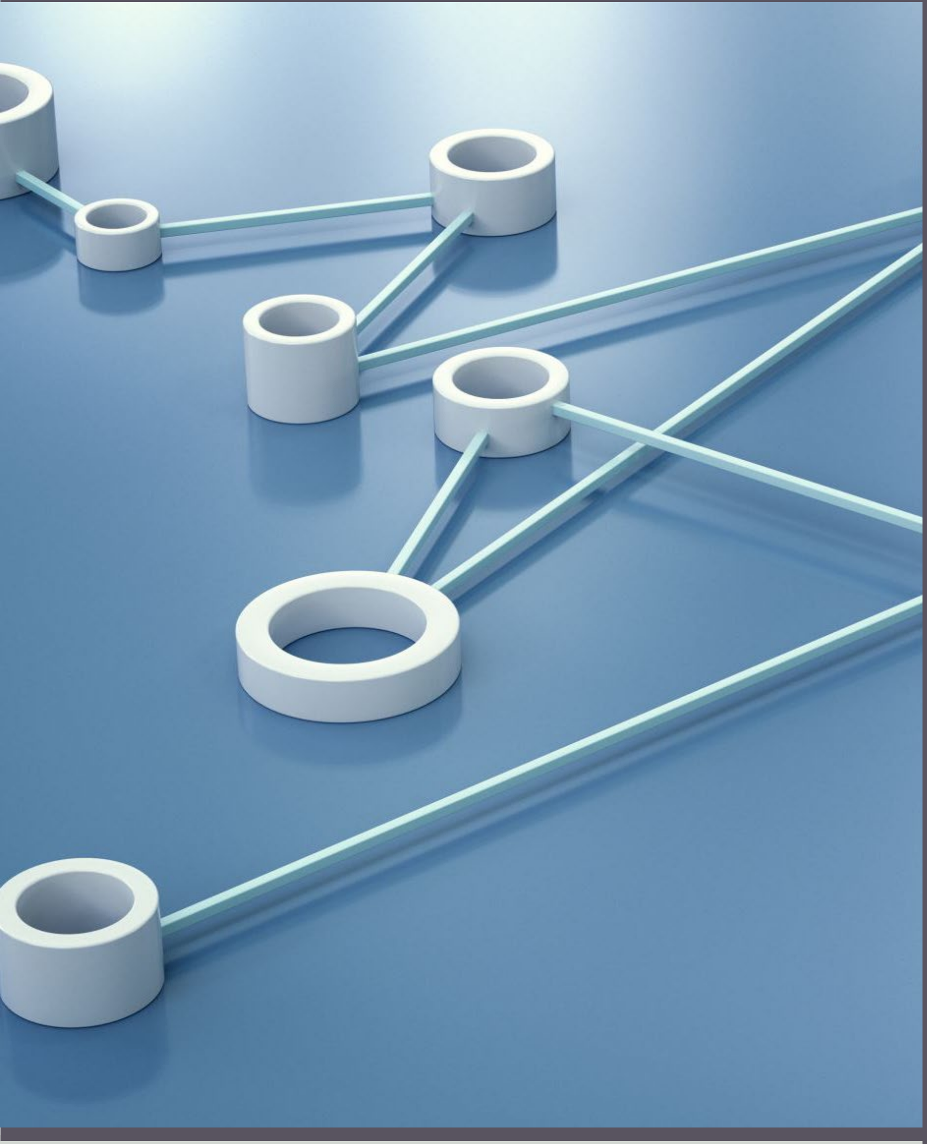
Strengths

Thorough experiments with diverse benchmarks and backbones validate the generality and effectiveness of MAS-GPT.

MAS-GPT significantly reduces the complexity of designing and deploying MAS for a wide range of tasks.

Weaknesses

Most the performance improvement seems coming from the GSM-H dataset. Improvements on other datasets are small and seems randomness.



Extension

Integrate Reinforcement : MAS-GPT generates an entire MAS system once per query, no iterative correction or adaptation.

Add reinforcement learning or self-improving feedback loops where MAS-GPT evaluates and revises its generated systems based on performance or feedback.

Self-evaluate

Self Correct

THANK YOU

Any questions?